Converting CryoSPARC Particles to RELION STAR Format using csparc2star.py

More details can be found on Daniel Asarnow's pyem github repository:

https://github.com/asarnow/pyem/wiki/Install-pyem-with-Miniconda

NOTES:

- Skip the Miniconda3 installation and go straight to the 'Create a new conda environment' section, as you will want to use the module version (details below).
- SBGrid (as of January 28, 2020) is using an older version of *csparc2star.py*. In the most recent version of *csparc2star.py* the default output is written in the new Relion 3.1 format. If you are not going to use Relion 3.1 you will want to append the --relion2 option flag at the end of the command. Note that the Relion 2.x format applies to Relion versions 3.x as well. Because of this, make sure that SBGrid is not loaded when you run the program. To prevent SBGrid from auto loading when you login to a terminal, create an empty file in your home directory:

  $ touch ~/.manual_sbgrid
- Particles exported from cryoSPARC 2D classification do not contain CTF parameters, so you will need to run *csparc2star.py* with the passthrough particles file that is output into the cryoSPARC output folder (details below). For the cryoSPARC programs that are run after 2D classification, the particle files are written with the CTF parameters.

Create a new conda environment

```
$ module load python-anaconda3/latest
$ conda create -n pyem
$ source activate pyem
$ conda activate pyem
$ conda install numpy scipy matplotlib seaborn numba pandas
$ conda install -c conda-forge pyfftw healpy pathos
```

Clone and install pyem

```
$ git clone https://github.com/asarnow/pyem.git
$ cd pyem
$ pip install --no-dependencies -e . --user
$ mkdir ~/bin
$ cd ~/bin
$ ln -s /path/to/pyem/*.py -t .
```

Running the pyem software

After successful completion of the pyem installation, the only requirements for invoking the program(s) are as follows:

```
$ module load python-anaconda3/latest
$ source activate pyem
```

Note that the pyem programs can be invoked from anywhere on the filesystem that you have permissions to read/write/execute files such as your home directory and project directory.

Some important csparc2star.py option flags

Here are a few of the most relevant option flags with descriptions on their use. You can generate the complete list with the --help option:

usage: csparc2star.py [input[input]] output [--option(s)]

- --help
  - List available option flags
- --micrograph-path MICROGRAPH_PATH
  - Replacement path for micrographs
- --copy-micrograph-coordinates COPY_MICROGRAPH_COORDINATES
  - Metadata file containing paths to both micrographs and particle coordinates (i.e. particles.star file from Relion extract job)
- --relion2, -r2
  - Produce output in Relion 2 format, which includes Relion 3.0.8. Note that the program by default produces the new Relion 3.1-star file format, which is not back compatible with older versions.

EXAMPLES:

The particles.star file that follows the --copy-micrograph-coordinates option flag was generated from a RELION extract particles job. Therefore, if you don't have a particles.star file you will need to determine proper input to point pyem to micrograph path.

Export particles from cryoSPARC ab initio reconstruction

```
$ csparc2star.py particles_selected.cs particles_selected.star -
-copy-micrograph-coordinates particles.star
```

Export particles from cryoSPARC 2D classification

```
$ csparc2star.py particles_selected.cs
passthrough_particles_selected.cs particles_selected.star --
copy-micrograph-coordinates particles.star --relion2
```